LEVEL 9: Return Book using Stack

1. Objective

Record recently returned books using LIFO. This introduces push, pop, peek, overflow and underflow.

2. Data Structure / Algorithm Used

Stack using Array

3. Problem Statement

Extend the Library Management System by implementing the return book using stack module. Complete this module independently before integrating it with previous levels.

4. Algorithm / Working

Push returned Book ID → top points to the latest return → peek/pop demonstrate LIFO.

5. C Program

```
#include <stdio.h>
#define MAX 10
int stack[MAX],top=-1;

void push(int id) {
    if(top==MAX-1) printf("Stack Overflow\n");
    else { stack[++top]=id; printf("Book %d pushed\n",id); }
}
void pop(void) {
    if(top==-1) printf("Stack Underflow\n");
    else printf("Popped Book: %d\n",stack[top--]);
}
void peek(void) {
    if(top==-1) printf("Stack Empty\n");
    else printf("Most recent return: %d\n",stack[top]);
}

int main() {
    int id;
    printf("Returned Book ID: "); scanf("%d",&id);
    push(id); peek();
    return 0;
}
```

6. Sample Input

Returned Book ID: 105

7. Sample Output

Book 105 pushed
Most recent return: 105

8. Complexity

Push/pop/peek: O(1). Stack storage: O(n).

9. Practice Exercises

- Push three returned books.
- Pop the latest return.
- Display the entire stack.

10. Related Viva Questions

- What is a stack?
- What does LIFO mean?
- What are push, pop and peek?
- What is overflow?
- What is the complexity of push and pop?